

Assignment 12

Consider $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$ and let $\psi: \mathbb{R} \rightarrow \mathbb{R}$ be given as

$$\psi(x) = x^2 \forall x \in \mathbb{R}$$

Then show that

a) For $f \in \mathcal{M}(\mathcal{B}(\mathbb{R}))$ show that

$$f \circ \psi \in \mathcal{L}^1(\lambda) \Leftrightarrow \int_0^\infty \frac{|f(x)|}{\sqrt{x}} \lambda(dx) < \infty$$

b) And in that case

$$\int_{\mathbb{R}} f(x^2) \lambda(dx) = \int_0^\infty \frac{f(x)}{\sqrt{x}} \lambda(dx)$$

a) " $\Leftarrow$ "

We want to show that

$$\int |f \circ \psi| d\lambda = \int_{\mathbb{R}} |f \circ \psi| \lambda(dx) < \infty$$

First we find the transformation measure

$$\begin{aligned} \lambda \circ \psi^{-1}((-\infty, b]) &= \lambda(\psi^{-1}((-\infty, b])) \\ \psi^{-1}((-\infty, b]) &= \begin{cases} \emptyset & \text{if } b < 0 \\ [-\sqrt{b}, \sqrt{b}] & \text{if } b \geq 0 \end{cases} \\ \lambda \circ \psi^{-1}((-\infty, b]) &= \begin{cases} 0 & \text{if } b < 0 \\ 2\sqrt{b} & \text{if } b \geq 0 \end{cases} = \mathbb{I}_{\{x \geq 0\}}(b) \cdot 2 \cdot \sqrt{b} \\ &= \int_{(-\infty, b]} t^{-\frac{1}{2}} \cdot \mathbb{I}_{\{x \geq 0\}}(t) \lambda(dt) \end{aligned}$$

By 1.1.3(B) this implies $\lambda \circ \psi^{-1} = \mu$

Where μ is a measure with density $t^{-\frac{1}{2}} \cdot \mathbb{I}_{\{x \geq 0\}}(t)$ with respect to the lebesgue measure

Then we have

$$\begin{aligned} \int_0^\infty \frac{|f(x)|}{\sqrt{x}} \lambda(dx) &= \int_{\mathbb{R}} \mathbb{I}_{\{x \geq 0\}} \cdot \frac{|f(x)|}{\sqrt{x}} \lambda(dx) \\ &= \int_{\mathbb{R}} |f(x)| \mu(dx) = \int_{\mathbb{R}} |f(x)| \lambda \circ \psi^{-1}(dx) \end{aligned}$$

And then

$$\int_0^\infty \frac{|f(x)|}{\sqrt{x}} \lambda(dx) < \infty \Rightarrow \int_{\mathbb{R}} |f(x)| \lambda \circ \psi^{-1}(dx) < \infty$$

And we're done

" $\Rightarrow$ "

By 11.1.4

$$f \in \mathcal{M}(\mathcal{B}(\mathbb{R})) \text{ and } f \circ \psi \in \mathcal{L}^1(\lambda) \\ \Rightarrow \int_{\mathbb{R}} |f(x)| \lambda \circ \psi^{-1}(dx) < \infty$$

And by the same argument as before

$$\int_{\mathbb{R}} |f(x)| \lambda \circ \psi^{-1}(dx) = \int_0^\infty \frac{|f(x)|}{\sqrt{x}} \lambda(dx) < \infty$$

b) By " $\Rightarrow$ " weve shown that $f \circ \psi \in \mathcal{L}^1(\lambda) \Rightarrow f \in \mathcal{L}^1(\lambda \circ \psi^{-1})$

And again by 11.1.4 we have

$$\int_0^\infty \frac{|f(x)|}{\sqrt{x}} \lambda(dx) = \int_{\mathbb{R}} f(\lambda \circ \psi^{-1})(dx) = \int_{\mathbb{R}} f \circ \psi \lambda(dx) = \int_{\mathbb{R}} f(x^2) \lambda(dx)$$